

Chapter 1

Snow products

1.1 Study area

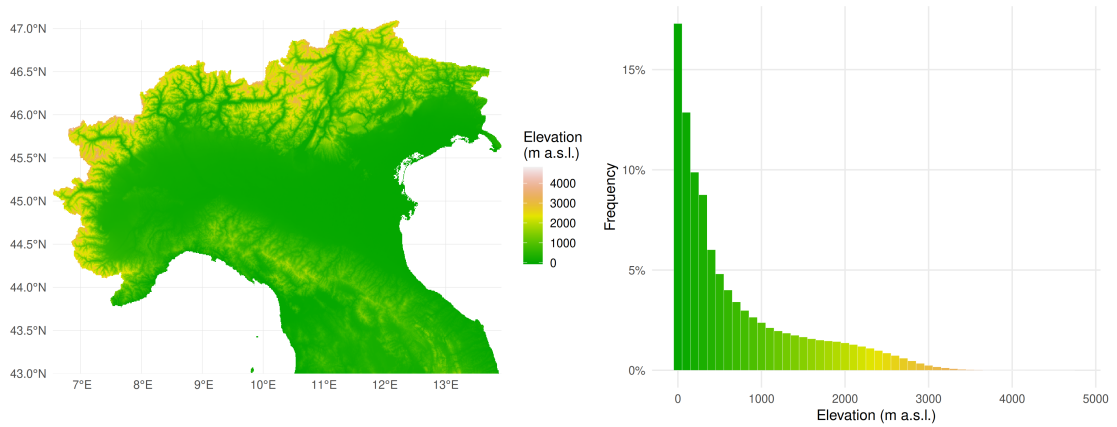


Figure 1.1: Topography and elevation distribution of the study area. The frequency histogram is based on 100 meters wide elevation bins. Elevation data were obtained from ALOS AW3D30 global digital elevation model (DEM).

1.2 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the whole Italian territory. It provides daily maps of snow water equivalent (SWE), snow depth, bulk snow density and liquid water content of the snowpack for a period spanning from September 1st, 2010 to August 31st, 2025. The spatial resolution is approximately 500 meters. IT-SNOW maps are the output of an operational chain developed by CIMA Research Foundation called S3M Italy.

1.2.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology-oriented cryospheric model that simulates seasonal snow evolution through time. S3M is a

raster-based model treating pixels as independent units, as it neglects mass and energy transport processes such as wind redistribution. The same set of ordinary differential equations (ODEs) is solved for each grid point by means of the Euler method: the hourly time-step can be adjusted by the user to meet specific application needs. All input and output variables are distributed.

S3M is specifically designed for large-scale, real-time application in order to be an useful tool in flood forecasting frameworks. This model achieves a high level of performance by focusing on key snow evolution drivers via precipitation-phase partitioning and a hybrid temperature and radiation-driven melt parametrization.

1.2.2 Results

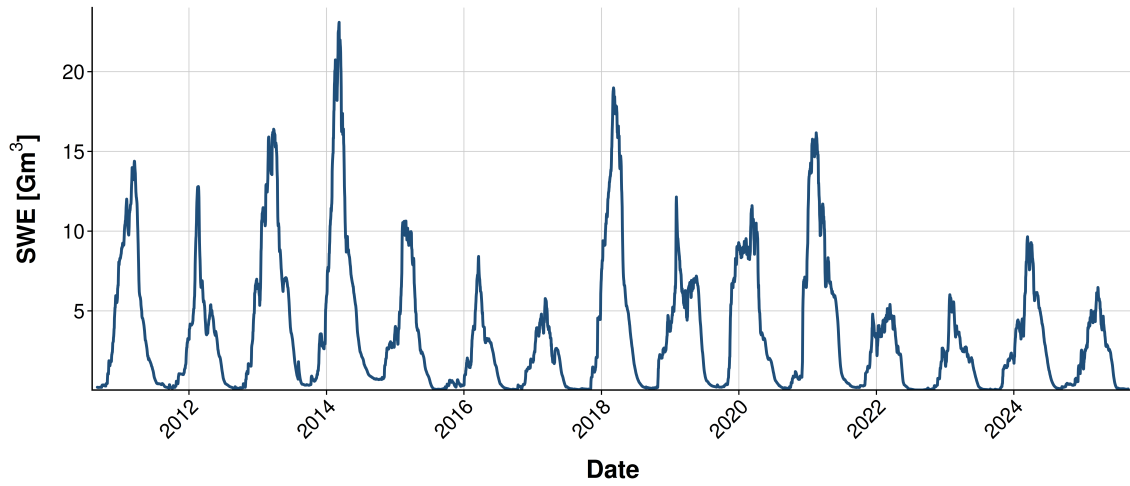


Figure 1.2: Total SWE evolution

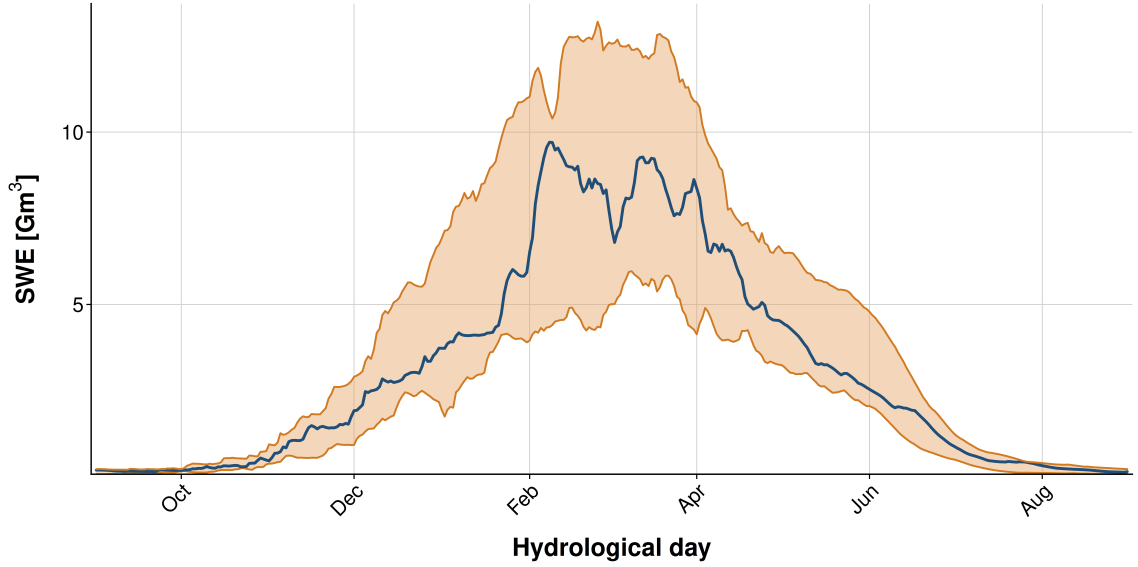


Figure 1.3: Total SWE climatologies.

1.3 MODIS

The role of Satellite Remote Sensing (SRS) in global climate monitoring has become increasingly critical, providing spatio-temporal observations across remote regions where conventional ground-based methods are lacking. In particular, those kinds of measurements are highly effective for mapping snow cover extent and duration. NASA’s Moderate Resolution Imaging Spectroradiometer (MODIS) is a major tool in this field: this 36-band sensor operates aboard two Earth Observing System (EOS) satellites, namely Terra (launched in December 1999) and Aqua (launched in May 2002). This tool has been used in order to assess snow cover duration (SCD) trends across the Greater Alpine Region (Fugazza et al. 2021) and to establish SCD climatologies across Europe. One of the latest studies on snow cover metric characteristics and trends focuses on the Italian territory (Manara et al. 2025), and therefore on the Italian Alps, which are the primary focus of this thesis.

1.3.1 Dataset Generation and Gap-Filling Strategy

In its latest version, this Italian snow cover dataset spans from 2000 to 2025 and consists of binary maps, where a value of zero represents to bare ground, whereas one is associated only to snow covered pixels. Available on a daily basis, these maps feature a spatial resolution of roughly 500 meters.

The starting point of the workline that led the authors to those maps are the raw modis datas. To correctly identify snow covered pixels advantage is taken on a

property of snow, which is that it has very high reflectance in the visible part of the electromagnetic spectrum, whereas it absorbs most of the shortwave radiation. This characteristics is highlighted by the normalized difference snow index (NDSI), which is defined as

where band 4 is in the visible range (in particular $0.545 - 0.565 \mu\text{m}$) and band 6 (or 7) is in the shortwave infrared (namely $1.628 - 1.672 \mu\text{m}$ for Terra and $2.105 - 2.155 \mu\text{m}$ for Aqua).

1.3.2 Results

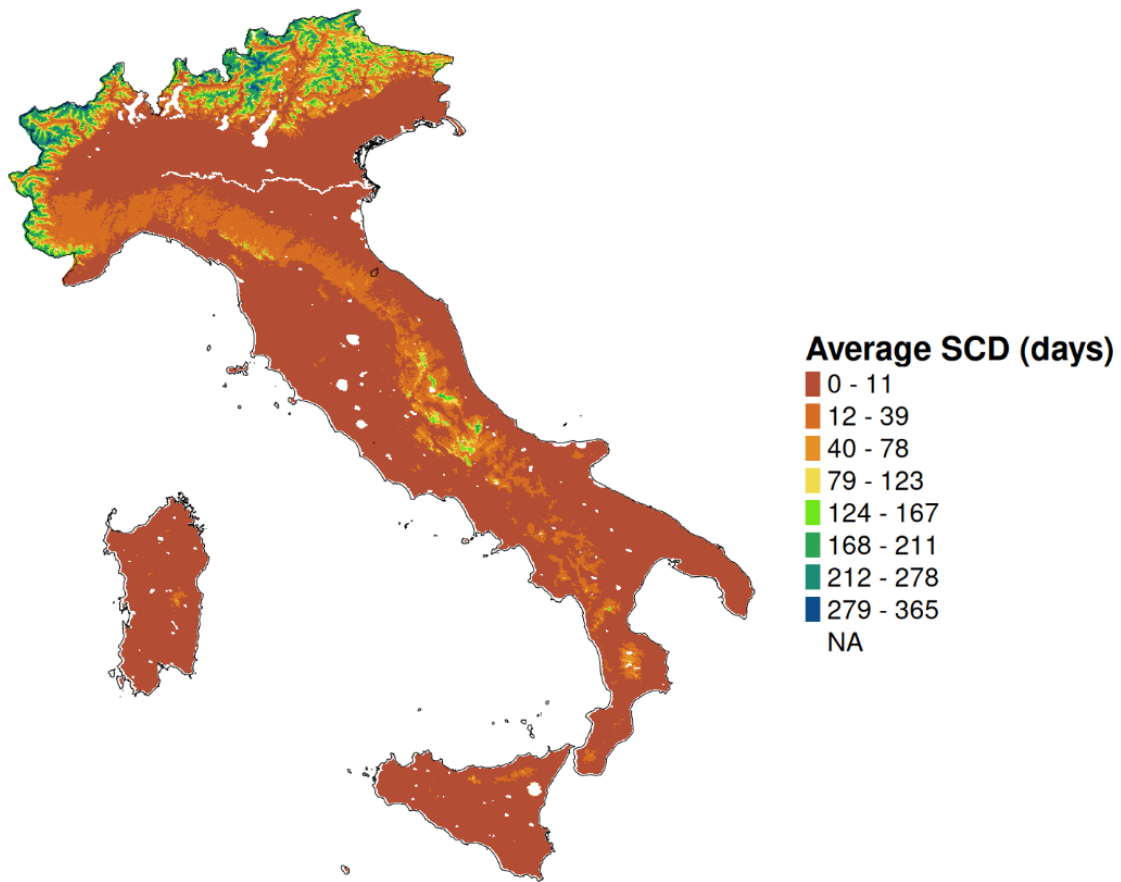


Figure 1.4: MODIS

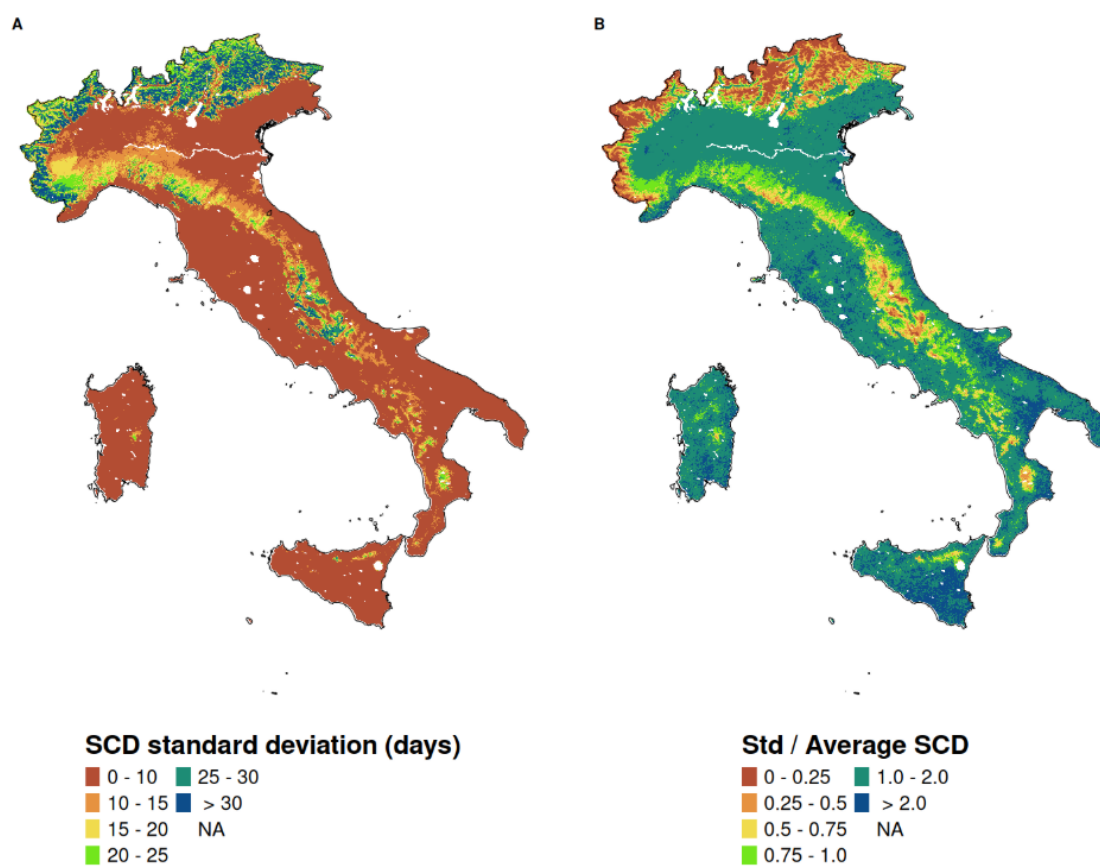


Figure 1.5: MODIS

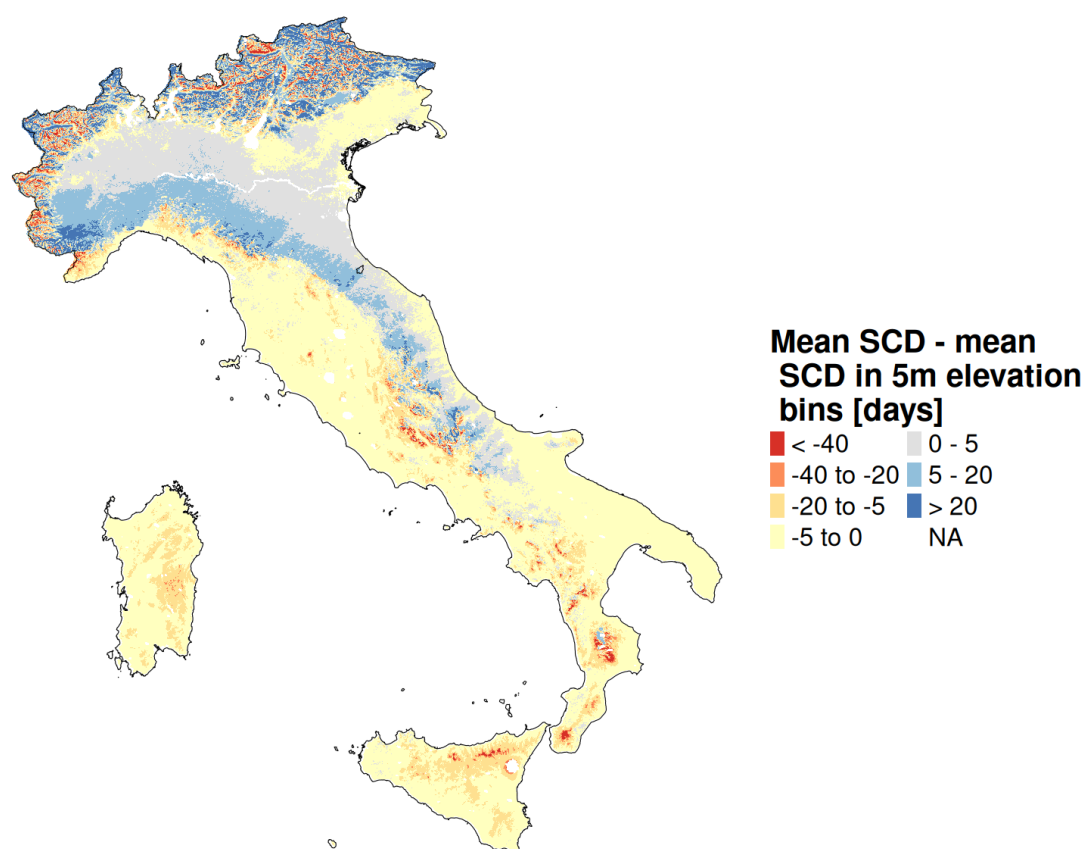


Figure 1.6: MODIS

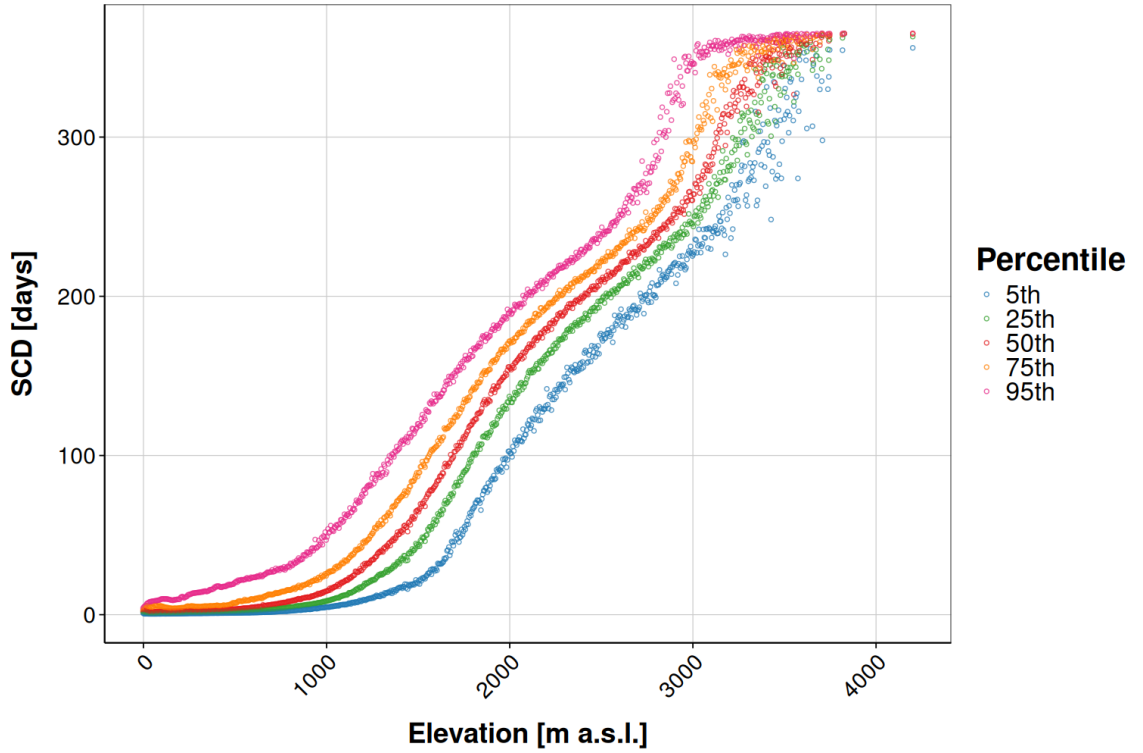


Figure 1.7: MODIS

1.4 Po Basin

This snow dataset focuses on the mountainous part of the Po River District, which is an over-regional entity resulting from the union of the Po River Basin with some smaller ones such as Conca and Reno basins. This territory, spanning eight regions of Northern Italy and parts of Switzerland and France, covers almost 87000 km^2 . Assessing the amount of water stored as snow in the highlands of this region is particularly important, because the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is a thirty years long SWE reconstruction, covering a period spanning from 1992 to 2021. For every hydrological year, SWE maps for the mountainous part of the Po River District are given on a daily basis during the period spanning from October 3rd to July 1st, which roughly corresponds to the period where snow coverage extends beside glacierized area. These SWE maps have a spatial resolution of $500 \times 500 \text{ m}^2$ and are created using a hybrid modeling approach "J-snow".

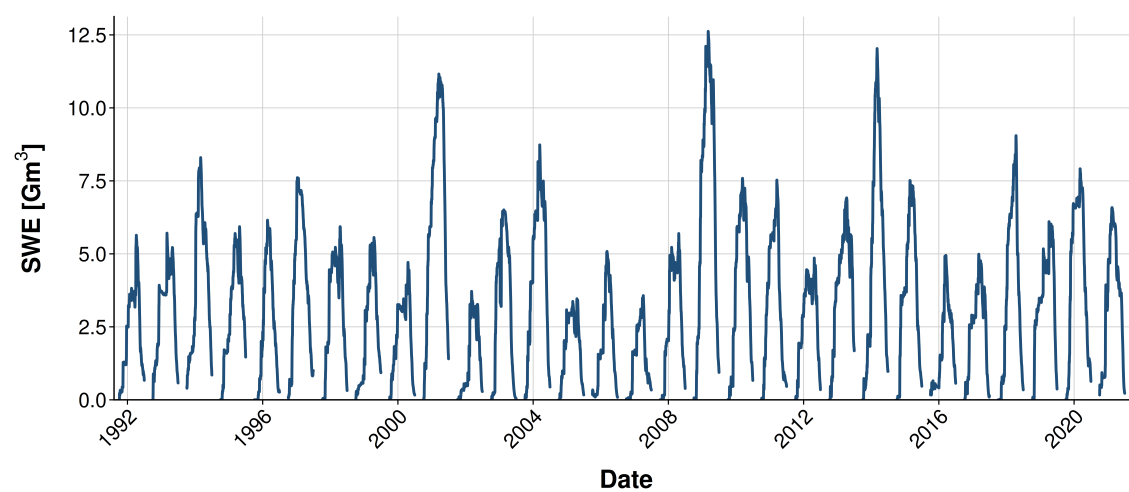


Figure 1.8: MODIS

Bibliography

Fugazza, Davide et al. (July 2021). “Snow Cover Variability in the Greater Alpine Region in the MODIS Era (2000–2019)”. In: *Remote Sensing* 13, p. 2945. DOI: [10.3390/rs13152945](https://doi.org/10.3390/rs13152945).

Manara, Veronica et al. (2025). “MODIS (2001-2022) snow cover variability over the Italian territory”. In: *Journal of Hydrology: Regional Studies* 62, p. 102895. ISSN: 2214-5818. DOI: <https://doi.org/10.1016/j.ejrh.2025.102895>.